

Aahan Jain

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Education

George Mason University

Master of Science in Computer Science

GPA: 3.73 / 4.0

Relevant Coursework: Machine Learning, Data Mining, Data Structures, Algorithm Design, Distributed Systems

Aug 2024 – May 2026

Fairfax, VA

Vishwakarma Institute of Technology

Bachelor of Technology in Computer Engineering

GPA: 3.37 / 4.0

Aug 2020 – May 2024

Pune, India

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL

Generative AI & LLM: Prompt Engineering, RAG, Agentic RAG, Multi-Agent Systems, LangGraph, LangChain, LlamaIndex

Frontend & Backend: React.js, Next.js, Django, FastAPI, REST APIs

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD, Git

Data & Analytics: MongoDB, PostgreSQL, pgvector, Pinecone, Vector Databases, ETL, Embedding Search

Machine Learning: XGBoost, PyTorch, LSTM, DistilBERT, Hugging Face, SHAP, scikit-learn, MLflow

Experience

Software Development Intern

Hexagon, India

Aug 2023 – April 2024

Hyderabad, India

- Shipped a production AI customer-support assistant on prompt-engineered **RAG** pipelines and **Azure OpenAI** to clear a growing backlog of repetitive tickets, fielding 500+ daily queries at 90% response accuracy.
- Orchestrated **LangChain** LLM workflows behind **CI/CD** automation on **AWS** to attack a slow 4-day resolution cycle, cutting resolution time to 2 days and stabilizing releases across 10+ client environments.
- Architected backend services in **Django** and **MongoDB** to remove latency bottlenecks in real-time AI query handling, lifting API response performance by 5%.

Research Experience

AI-Accelerated Software Engineering (DocuMind)

George Mason University — CS 691

Sept 2025 – Dec 2025

[GitHub](#)

- Engineered DocuMind, a 1,000+ line **LangGraph** multi-agent **Agentic RAG** platform that ingests **GitHub** repositories and parses multiple languages into structural metadata to solve how developers explore and document large, unfamiliar codebases.
- Implemented retrieval over **LlamaIndex** embeddings in **Pinecone** with semantic search, metadata filtering, and re-ranking, scoring faithfulness and relevance through **RAGAS** and **LangSmith** to keep **FastAPI** responses grounded and hallucination-resistant.
- Generated technical documentation and architecture diagrams through prompt-engineered **LangGraph** agents, cutting new-developer onboarding time by 30%.

Projects

AI Decision Intelligence Platform for Small Businesses

Mar 2026 – Present

- Designed an AI decision-intelligence platform that tells small-business owners what to do next and why, orchestrating context, forecasting, recommendation, and explanation through a six-agent **LangGraph** workflow.
- Modeled demand, pricing, and sentiment with an **XGBoost**, **PyTorch LSTM**, and **DistilBERT** ensemble, adding **SHAP** explanations and a **pgvector RAG** memory that grounds each recommendation in the business's past outcomes.
- Automated a feedback-driven retraining loop tracked in **MLflow** with **Prometheus/Grafana** monitoring, served via **FastAPI** behind a **React/Next.js** app on **Docker/Kubernetes**.

Customer Recommendation System Through Transaction History [\[GitHub\]](#)

Jan 2025 – May 2025

- Built a decisioning system that scores each customer's likely next purchase and selects the highest-value offer to surface, pairing behavior-driven personalization with **GPT**-written alerts to replace generic, poorly timed promotions.
- Trained a **scikit-learn Random Forest** classifier on quantile buckets of days-to-next-purchase with cross-validation, engineering features over 22,500 synthetic transactions across 150 customers to time and rank vendor-matched offers.
- Delivered **GPT**-generated deal-alert copy with a graceful fallback through a **Django** REST API and **Next.js** dashboard, secured with **JWT** auth and packaged via **Docker** Compose.

Product Experimentation & A/B Testing Platform [\[GitHub\]](#)

June 2025 – Sept 2025

- Developed a **Python** experimentation framework that simulates events, runs an **ETL** pipeline into variant/cohort metrics, and sizes conversion lifts with **SciPy/statsmodels** z-tests and multiple-testing correction.
- Quantified prompt quality across 5 LLM variants over 26,000 evaluations with paired t-tests; modeled conversion via **scikit-learn logistic regression** (ROC-AUC).